Heathkit HW-101 Transceiver — Restoration Plan

Rev. 1 — field sources, Heath service bulletins, W5RKL and Latta notes
(manual cross-check pending)

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Safety. The HW-101 runs on **~700–800 V DC** plate supply from the HP-23 power supply. That voltage stays on the filter capacitors after the supply is unplugged. Discharge with a grounded bleeder stick before every hands-in session, and never operate the final cage with its cover off. Work one-handed when the set is live.

1. Source documents

Document	Where	Use
Heath HW-101 Assembly & Operation manual (245 pp.)	Bill Latta's HW-101 site, frostburg.edu	Primary reference: parts lists, voltage charts, alignment, schematic. *Scan only; cross-check pending (§11).*
HW-101 Service Bulletins, HW-101-1 … -85 (1971–79)	Latta site; text of -1 to -55 on the Heath list (Jul 1997)	Factory fixes. The bulletin numbers below come from these.
HP-23 power supply manual	Latta site	Required companion supply.
W5RKL, *Heathkit HW-101 and SB-10x Restoration and Troubleshooting*	w5rkl.com	First power-up checklist, bias/neutralizing, alignment checks.
Latta (Frostburg) HW-101 restoration, alignment, HP-23 restoration pages	frostburg.edu	Documented restoration with measured failures.
WB8JKR, *Optimizing the HW-101 / SB-100–102*	mikeyancey.com	Meter-zero, carrier-crystal, receive/transmit improvements.
K4JPN, *Improve Your HW-101*; **KM5Z** refurb notes	nostalgickitscentral.com, pestingers.net	Tube-brand problems, T/R relay, pilot lamps.
Heath list archive / Heathkit-Radios (groups.io)	puck.nether.net, groups.io	Field experience.

2. Transceiver at a glance

2.1 Key facts

Item	Value
Built	1970–1983; 30,000–40,000 kits. Nearly identical to the SB-102.
Bands	80, 40, 20, 15, 10 m (eight 500 kHz segments, 3.5–29.5 MHz)
Modes	SSB (USB/LSB), CW
IF	3395 kHz; 1st conversion 8.395–8.895 MHz bandpass (T202)
Filters	2.1 kHz SSB (supplied); optional 400 Hz CW, front-panel switched
Power	180 W PEP input SSB, 170 W CW; about 100 W out (80 W on 10 m)
VFO	Solid-state: **MPF-105 FET** oscillator, **2N3393** (Q942) regulator, V20 6AU6 buffer

Item	Value
Supply	External **HP-23** (A/B/C) or PS-23 (AC), HP-13 (mobile)
Pilot lamps	**#47** (part of the 12.6 V filament string balance; bulletin HW-101-20)

2.2 Tube complement

The V numbers below are assembled from the field sources and bulletins. **Verify them against the manual's tube chart before ordering.**

V	Tube	Function	Known issues (source)
V1	6EA8	Speech amp / cathode follower	GE 6EA8s oscillate (~65 kHz) → VOX chatter, no carrier null (HW-101-21, -31). 6GH8A/6CQ8 are used as substitutes.
V2	6AU6	Carrier isolation amp	—
V3	6AU6	1st IF amp	Leakage → S-meter drift (HW-101-34). El-Menco brand draws excess current → can't zero meter.
V4	6AU6	2nd IF amp	Brand-sensitive: GE / IEC OK; some Sylvania/RCA/Westinghouse didn't work (HW-101-46).
V5	6EA8	1st transmit mixer	Check seating.
V6	6CB6	2nd transmit mixer	Under shield; shield contact matters.
V7	6CL6	Driver	Under shield.
V8, V9	**6146A / 6146B**	Finals (class AB1)	1978: Heath said use 6146A only (HW-101-18, -39). 1979: GE 6146B approved (HW-101-49). **Re-neutralize after any change.**
V10	6HS6	Receiver RF amp	**High failure rate.** Leakage or shorts → poor AGC, fast S-meter decay, lost sensitivity, "static" on S-meter (HW-101-24). Run at or above ratings. A 6AU6 is a drop-in substitute.
V11	6HS6	1st receive mixer	Same as V10. Some 6HS6s had swapped pinouts (HW-101-12).
V12	6EA8	2nd receive mixer / relay amp	Defective (IEC) tubes hold relays in after TX (HW-101-23); El-Menco may oscillate (HW-101-17).
V13	6BN8	Product detector / AVC	Weak 6BN8s found on test.
V14	6GW8	Audio amp / output	Runs hot.
V15	6EA8	CW sidetone osc / amp	Shorts → sidetone problems or continuous keying (HW-101-74). A tube shield cures bleed-through.
V16	12AU7	Carrier (sideband) oscillator	Plate resistors R6/R7 drift (§3.2).
V17	12AT7	VOX amp / 100 kHz calibrator	—
V18	0A2	150 V regulator (VFO)	Flicker on CW points to an HP-23 on the 250 V tap or a bad C5.
V19	12AT7	Heterodyne osc / cathode follower	Repeated failure: change R212 (HW-101-2).
V20	6AU6	VFO buffer amp	Feeds the FET's ~8 V regulator; a poor tube → drift. El-Menco → CW chirp.

Brand note (K4JPN, W5RKL): El-Menco 6AU6, 6HS6, 6EA8, 12AT7, 12AU7 and 6CL6 tubes

are repeatedly implicated. Keep spares of other brands.

2.3 Semiconductors

The HW-101 has 6 germanium diodes, 9 silicon diodes, 1 zener, the MPF-105 FET and the 2N3393. Named in the bulletins:

- **CR1–CR4** (modulator): germanium; bulletin HW-101-13 upgraded them to hot-carrier diodes (56-87).
- **CR201** (calibrator), **CR901** (relative power), **CR941**: germanium 1N191-type (56-26).
- **D201**: VOX. Leakage here kills VOX sensitivity (HW-101-6).
- **D907**: orientation affects 15/10 m receive sensitivity (HW-101-11).

3. What fails, ranked

3.1 Capacitors

Electrolytics in the transceiver (Latta's measured list; W5RKL gives 450 V/50 V ratings for the SB versions):

Ref	Value	Role	Replace with
C2	10 µF 15 V	Speech amp cathode bypass (modulator board)	10 µF 50 V
C12	20 µF 350 V	Speech amp B+ decoupling	22 µF 450 V
C212	10 µF 15 V	VOX amp cathode bypass (bandpass board)	10 µF 50 V
C304	20 µF 350 V	Audio output B+ decoupling (audio board)	22 µF 450 V
C906	10 µF 150 V	Bias filter	10 µF 250 V; **negative supply, so check polarity**

Keep the same capacitance in audio and speech circuits so the frequency response doesn't change. A higher voltage rating is fine. The main filter capacitors are in the HP-23 (§3.6).

Other capacitor	Likelihood	Symptom	Action
C946 (4700 pF, VFO)	Medium	Warble when tapped	Glue its top to the VFO wall (HW-101-42).
C947 (56 pF, VFO shift)	Medium	Intermittent USB/LSB shift	47 pF NPO (HW-101-19).
C213 (.2 µF, VOX delay)	Medium	Short VOX hold, VOX trouble	Leakage-test; .47 µF (HW-101-63).
C701, C801 (680 pF 300 V)	Medium	R940 overheats	680 pF 500 V (HW-101-75).
Silver micas in carrier/BFO (C17 etc.)	Low–medium	Weak or uneven carrier	Test; WB8JKR changes C17 from 12 pF to 100 pF.

Other capacitor	Likelihood	Symptom	Action
Ceramic discs	Low	—	Spot-test bypasses.

3.2 Resistors

These are 1/2–2 W carbon composition parts, ~50 years old. They drift high, worst where they run hot.

Ref	Value	Failure / symptom	Fix (source)
R106	22 kΩ 1 W (IF board)	Heats and changes → **S-meter zero drift**	22 kΩ 2 W wirewound (WB8JKR); K4JPN uses 33 kΩ 2 W. Four other connections share its pad, so keep them all.
R107	100 kΩ 1/2 W	S-meter drift with heat	100 kΩ 1 W (HW-101-10).
R6, R7	33 kΩ (carrier osc V16 plates)	One measured **330 kΩ**, another 42 kΩ → low USB/CW gain and output	Replace with 1 W (HW-101-84); WB8JKR uses 27 kΩ to raise carrier level.
R1	100 kΩ (speech amp)	Heat damage	1 W (HW-101-84); also check R215 100 kΩ and R316 22 kΩ.
R117	3.3 kΩ (AVC)	Open → AVC and S-meter decay too fast	Replace; route the AVC wire ends away from it (HW-101-55).
R915	3.3 MΩ	Open → no USB/LSB output (TUNE OK), ALC upset	Replace (HW-101-69).
R940, R904	100 Ω	Overheat or short to the driver shield	Sleeve the leads, dress away from the shield (HW-101-37, -50).
R701	1 kΩ (driver grid switch board)	Burned, from a bandswitch lug shorting to the shield cover	Replace and bend the lug clear (W5RKL).
R212	220 Ω	Repeated heterodyne-oscillator tube failure	330 Ω 1/2 W (HW-101-2).
R945, R947	VFO	Low VCO output; VFO stops at the high end	Per HW-101-52, -80 (§4.6).
R202	10 kΩ	*Early units only.* Limits drive to V5	Jumper it (HW-101-14). Latta shunted it with wire to keep the early-run marker.
10 kΩ plate-supply resistors (driver / audio)	—	Drift high → low output or weak audio	Measure and replace.

Method: lift one lead, measure, and replace parts more than 10–20% out of tolerance. Use

carbon comp or carbon film in RF and oscillator circuits.

3.3 Tubes, by risk

1. **6HS6 (V10, V11):** the top HW-101 receive killer. Try a 6AU6 as a test substitute.
2. **6146 finals (V8, V9):** emission, gas, mismatched pairs. Only service in the set proves a final.
3. **6EA8 (V1, V12, V15):** oscillation, relay hang-up, sidetone faults.
4. **6AU6 (V3, V4, V20):** brand-sensitive; S-meter zero and chirp problems.
5. **12AT7 (V19), 12AU7 (V16), 6BN8 (V13), 6GW8 (V14), 0A2 (V18).**

Put each tube back in the socket it came from. Alignment is tube-dependent.

3.4 Mechanical, switches, relays

Item	Symptom	Fix (source)
RL2 contacts (carbon)	Chirp, slow receive recovery, no audio after TX. +300 V can leak onto the bias/AGC lines.	Clean contacts 2 and 10 with typing paper; clean carbon from the relay base (HW-101-32). Covers pull straight up.
RL1 / RL2 general	Chatter, sticking	Card-strip cleaning; VOX chatter bulletins -31, -41, -57.
Bandswitch wafers	Dead or partial bands	DeoxIT and rock-test. **Check wafer notch orientation.** One list unit had 3 of 5 wafers reversed from the original build.
L904 → bandswitch wires (5)	Short if run without a load	Sleeve or replace (K4JPN).
SSB/CW filter slide switch (by RF gain)	Low TX output **and** low RX sensitivity	Clean (HW-101-53).
Rotary switch detents	Broken detent	Newer detent 266-1116 (HW-101-66, -71).
Driver preselector / load rubber belts	Preselector won't track, slips	Silicone belts, 50 mm ID / 56 mm OD / 3 mm. Same belt for SB-100/101/102 and HW-100/101.
Preselector tuning drums	Loose on the shaft	Grind off the old solder, epoxy-bond (Latta).
Preselector caps (26-122)	Stiff; corroded pins	Clean or tin the pins; replace if they bind (HW-101-29, -81).
Loading cap turns with plate	—	Grommet 73-3 or fiber washer on the shaft (HW-101-30, -64).
VFO vernier / stop stud	Erratic VFO	Braid from the stop stud to a frame lug (HW-101-27).
VFO tune slips	—	Nylon and spring washers behind the loading pulley (HW-101-78).
VFO spiral and follower	Stiff dial	Clean the groove, silicone-lube, oil the pawls; regrease the capacitor if stiff (W5RKL).
Coil covers	Can't reach driver/het coils	Enlarge holes with a Dremel #194 bit (Latta).

Item	Symptom	Fix (source)
RF driver board screws / lockwashers	Self-oscillation	Tighten and fit lockwashers (HW-101-9).
Carrier null pot (10-147)	Fails, noisy	Fiber washer and ground wire (HW-101-3); or a 200 Ω trimpot (KM5Z).
Pilot lamps	Low output, S-meter drift	**#47**, never #44; never run with a lamp out (HW-101-20).

3.5 Calibrator and crystals

- **Y201 100 kHz crystal:** a choppy output causes spurs and raised noise, and can trip anti-VOX so the rig sits in transmit (HW-101-25; Latta). Replace (404-43).
- **Band crystals** are band frequency + 8.895 MHz; for example 29.5 MHz uses 38.395 MHz. They're shared with the SB-102. Latta's 29.5 MHz crystal was dead.
- **Carrier crystals** have wide tolerance. WB8JKR measured LSB 3393.8, USB 3395.9 and CW 3395.17 kHz, and pulled them back to 1.4–1.6 kHz either side of the filter center. CW output went from 50 W to 110 W.

3.6 HP-23 power supply (do this first)

Item	Original	Failure	Replace with
C1, C2 (HV, in series)	125 µF 450 V	Low capacitance (Latta)	150 µF 500 V
C3, C4 (B+)	125 µF	Aging	150 µF 500 V
C5 (B+ filter)	40 µF 450 V	**Open → tone on CW, relay chatter** (HW-101-73)	47 µF 450 V
C6, C7 (bias)	40 µF 150 V	**Both open** (Latta)	47 µF 160 V+
R1–R5 (HV bleeders)	100 kΩ 2 W	Measured ~150 kΩ	100 kΩ 2 W metal film
R7 / R8 (bias)	1 kΩ / 27 kΩ	Drift; R8 was 1 W	2 W metal film
D1–D7	1N2071	Usually OK	1N4007
Line cord	2-wire	Safety	18/3 grounded

Outputs (11-pin):

Pin	Output
1	−130 V bias
2, 6	12.6 VAC filaments
3	B+ 250/300 V
4	HV ≈ 700 V loaded (~800 V unloaded)
7	Ground
9–10	Power-switch loop

The B+ switch must be on 300 V for the HW-101. On 250 V you get low TX power, VOX trouble and V18 flicker on CW. The older 250/OFF/300 switch is easy to bump.

Restored-board kits (for example HP-23RL) replace most of these parts in one step.

4. Restoration plan

4.1 Phase 0: Assess cold

- Photograph everything. Note mods such as RIT added to V20, R202 present or jumpered, BNC/SO-239 output and non-stock tubes.
- Pull and label every tube by socket. Test them, and note the brands (El-Menco, GE 6EA8, IEC).
- **Check bandswitch wafer orientation** against the manual's pictorials.
- Inspect: R940/R701/R904 (scorching), the final cage, the RF choke (fragile wire), L904 wires, RL1/RL2 bases, belts and drums.
- Resistance-check the HW-101 B+, HV and bias lines to chassis, using the manual's chart, before connecting a supply.

4.2 Phase 1: HP-23 first

Recap the HP-23 as in §3.6. Bring it up on a variac at 10 → 25 → 50 → 75 → 100%, unloaded, with a dummy load resistor on each output. Confirm about −130 V bias, 300 V B+ (switch!) and about 800 V HV unloaded. Check its primary (plug, fuse, switch) with an ohmmeter first.

4.3 Phase 2: Clean

- Vacuum and brush; gentle compressed air.
- Use cleaner on a rag or swab, never sprayed directly. Test naphtha on a hidden spot first.
- Final cage: finals out; pipe-cleaner between the tuning-cap plates.
- Front panel and windows: mild cleaner only; plastic dulls easily.
- DeoxIT on every rotary switch (including the bandswitch), pot and the SSB/CW slide switch.

4.4 Phase 3: Transceiver recap and resistor survey

1. Replace C2, C12, C212, C304 and C906 (C906 negative polarity).
2. Replace the drift-prone resistors in §3.2: R106 (2 W), R107 (1 W), R6/R7 (1 W), R1 (1 W). Check R117, R915, R212, R940, R701 and R904.
3. Lift-and-measure survey of the remaining resistors, worst-heat areas first (audio, VFO buffer, driver, modulator).
4. Clean RL2 contacts and base, and RL1.
5. Fit #47 pilot lamps.

4.5 Phase 4: Mechanical

Fit silicone preselector/load belts, epoxy loose drums, ground the VFO stop stud, glue C946, and enlarge coil-cover holes if needed.

4.6 Phase 5: VFO

The VFO is solid-state with a tube buffer; it is not an LMO.

- Drift: try several V20 6AU6s (use the least-drifting). Check V20 pins 5/6 at about 149 V with VFO B+ at 150 V (V18 0A2). Suspect Q942 2N3393, which regulates the FET's ~8 V.
- Original-coil drift: coil 40-810 was superseded by 40-1976 (HW-101-1).
- Stops at high end: R947 1 kΩ → 470 Ω and a diode from Q941 gate to ground (HW-101-52), or lower T941 output to ~2 V RF (HW-101-77).
- Tracking: use the **lower** of the two coil-slug peaks (HW-101-45).
- Remove any RIT mod wired to the buffer tube (field report: it caused one-directional shift).

4.7 Phase 6: First power-up

1. HP-23 on a variac. **Dummy load (50 Ω, not a light bulb) on the antenna jack.** Speaker connected.
2. Raise slowly; watch for heat or smell. Both pilot lamps must light; they balance the filament string.
3. Log voltages against the manual's voltage chart.
4. Check receive on every band with the calibrator.
5. **Set final bias before any transmit** (§4.8 T1).

4.8 Phase 7: Alignment

Order matters: **receiver first**, because the transmitter shares receiver circuits. Warm up for 30 minutes. Use the plastic tool. **Never adjust the sealed 6.8 MHz trap.** T202 (bandpass) is factory set; touch it only with a sweep setup (HW-101-56).

Receiver

1. **S-meter zero:** 80 m, RF gain fully CW, METER on ALC. Adjust ZERO ADJ (IF board, behind the FREQ CONTROL switch).
2. **Heterodyne oscillator coils:** use an 11 MΩ meter (an analog meter must be ≥ 20 kΩ/V) at test point TP near V19 on the bandpass board.
   - Target: as high as possible between −0.5 and −2.0 V, **never above −2.0 V**.
   - Approach the setting from the slow-changing side of the peak.
   - Five coils are adjusted from the bottom (21, 7, 29.5, 29.0, 28) and three from the top (3.5, 14, 28.5). If TP reads 0 V, measure at the other end of the 100 kΩ resistor.
3. **IF:** calibrator at dial 400. Peak T201, T102 (top and bottom slugs), then T103. Repeat. WB8JKR peaks T102 for maximum transmit drive if there are two peaks.
4. **VFO:**
   - Set the dial with the Zero-Set button and slip clutch.
   - Dial 500 = 5000 kHz (VFO coil); dial 0 = 5500 kHz (side trimmers). Iterate.
   - The dial reads inverted: 5.000 MHz at 500.
   - Without a counter, verify against a known signal; the error can reach 100 kHz.
5. **Driver grid and plate coils:** in the order 3.5 (3.7) → 29 (29.2) → 21 (21.2) → 14 (14.2) → 7 (7.2). Set the driver preselector to match each band and peak the S-meter. On 40 m, the preselector may need to sit near 1 o'clock (HW-101-47).

Receiver benchmark (W5RKL): calibrator about **S9+20 at 3700 kHz**, falling to about **S6 at 29.2 MHz**.

Transmitter (50 Ω dummy load; short key-downs; let the finals cool ≥ 30–60 s between tries)

1. **Final bias:**
   - Set up: PTT, LSB, no drive (MIC/CW level at zero), METER on PLATE.
   - Adjust BIAS for the manual's resting plate current: the **50 mA mark/dot** (W5RKL); Latta reads it as "3" on his scale.
   - **Confirm the procedure in the manual.** Wrong bias for more than a few seconds can ruin the finals.
2. **Carrier isolation amp T1:**
   - Use the REL PWR meter in TUNE, keeping output ≤ 6.
   - Peak the preselector and FINAL.
   - Reduce MIC/CW level to ≤ 3 and peak T1. Repeat once.
3. **Preliminary neutralization (3.5 MHz):** peak RF output and plate-current dip must fall at the **same FINAL setting**. Adjust the cage-front neutralizing capacitor 1/16–1/8 turn at a time. If the two points move apart, go the other way.
4. **Carrier null (3700 kHz):**
   - Set up: LSB, PTT, MIC/CW fully CCW.
   - Null with the CARRIER NULL control and capacitor, then repeat on USB.

- Aim for equal residual carrier on both sidebands, about ¼ V on the REL PWR meter or less. Spec is −45 dB.
- If C14 bottoms out, see HW-101-28. If suppression is poor, change R9 from 1 kΩ to 390 Ω (HW-101-38).

5. **Final heterodyne-coil touch-up:** TUNE, REL PWR 3–9. Peak the 3.5 coil, then each band's coil.
6. **Driver neutralization (21.2 MHz):** set the white wire to the 6CL6 plate (hole W) so the preselector peaks smoothly. Shorten it to about 1/8 in. if final neutralization is hard (K4JPN). On 15 m, bend it flat if the driver oscillates (HW-101-79).
7. **Final neutralization touch-up at 14.2 MHz.** It is more sensitive there.
8. **Calibrator:** zero-beat a harmonic against WWV at 10 or 15 MHz with the CAL CRYSTAL trimmer, during an unmodulated period.
9. **VFO shift:** dial 200 on 3.5 MHz, CAL on. Zero-beat in USB, switch to LSB without touching the dial, and adjust SHIFT ADJUST on the VFO. Iterate.
10. **Loading:** dip with FINAL, load to about **40** on the HW-101 plate scale (≈ 225–250 mA). Re-neutralize if maximum REL PWR and the dip don't coincide.

Changing finals, tube brands or layout means re-neutralizing.

4.9 Phase 8: Optional improvements (reversible)

Mod	Change (source)
Stable meter zero on TX	10 MΩ from meter-zero pot pin 3 to T102 pin 2 (WB8JKR)
Carrier crystals on frequency	Series/parallel trim caps; LSB 3393.5, USB 3396.3, CW 3395.4 kHz (WB8JKR)
More BFO injection	C17 12 → 100 pF; R6/R7 → 27 kΩ (WB8JKR)
Less raspy audio	C119 500 pF → .001 µF; C306 → .01 µF (WB8JKR)
Better mic low end	C11 .001 → .01 µF (WB8JKR)
Faster CW relay pull-in	R328 470 kΩ → 1 kΩ (WB8JKR)
Sidetone bleed in receive	.001 µF RL1 pin 1 to ground; R326 → 2.2 MΩ (WB8JKR) or 3.3 MΩ (HW-101-16)
Sidetone volume control	HW-101-70
Birdies	.01 µF from the common filament point to ground (bandpass board)
Low-Z headphones	WB8JKR rewiring; .01 µF across phone jack pins 2–3 (K4JPN)
RIT	QST Mar 1969 (SB-101 design)
Avoid	Replacing the RCA antenna jack with SO-239. Use an adapter (Latta).

5. Bulletin quick-reference (symptom → bulletin)

Symptom	Bulletins
S-meter drift / won't zero	-10, -20, -34, -67; R106
AGC poor, fast S-meter decay	-24, -55
VFO drift, warble, erratic, stops high	-1, -2, -19, -27, -42, -45, -52, -77, -80
Relays hang / chatter / chirp	-23, -31, -32, -41, -57
Low TX output or RX sensitivity	-11, -43, -46, -53, -56
Carrier suppression / null	-3, -28, -38

Symptom	Bulletins
Finals / neutralizing / RF choke heat	-5, -18, -39, -49, -59, -61
Calibrator spurs	-25
R940 / R701 overheating	-37, -50, -75
Sidetone	-16, -65, -70, -74

6. Operation notes

- Make every transmit adjustment into a **50 Ω dummy load**.
- Keep the HP-23 switch on **300 V**.
- Never transmit without a load; the L904 wiring and finals suffer.
- Tune up at reduced drive and keep key-down periods short.
- Use the calibrator and zero-set before netting.

7. Tips and tricks

- Label tubes by socket. Tube brand matters in this rig (6AU6, 6HS6, 6EA8).
- Photograph wafer and rotor orientation before touching the bandswitch.
- Grid-dip the heterodyne coils to find dead band crystals quickly (crystal = band + 8.895 MHz).
- Spare tubes worth keeping: 6HS6 ×2, 6EA8, 6AU6 (non-El-Menco), 12AT7, a matched 6146 pair.
- Keep #47 lamps, silicone belts and a 100 kHz crystal on hand.

8. Maintenance schedule

Interval	Task
Each session	30 min warm-up; calibrator zero-set.
6 months	Exercise switches and pots; check S-meter zero.
Yearly	Bias check; carrier null check; receive benchmark; clean relay contacts if chirp or slow recovery appears.
2–3 years	Neutralization check (peak and dip coincide); retighten board and driver screws; DeoxIT switches.
5 years	Test all tubes (6HS6 and finals first); inspect belts; HP-23 voltage check and capacitor ESR/leakage.
~15–20 years	Re-check electrolytics in both units.

9. Parts sources

- Mouser / Digi-Key: electrolytics, 2 W metal-film and wirewound resistors, NPO caps, 1N4007.
- HP-23 recap and rebuild boards (HP-23RL and similar).
- Silicone preselector belts (50/56/3 mm) from Heathkit vendors and eBay.
- Heathkit-Radios (groups.io): crystals, 6HS6s, donor boards.

10. Sources

- Heath, *HW-101 Assembly & Operation Manual*; *HW-101 Service Bulletins* 1971–79 (Latta site, frostburg.edu).
- Heath list (puck.nether.net), Jul 1997: transcribed bulletins HW-101-1 to -55.
- W5RKL, *Heathkit HW-101 and SB-10x Restoration and Troubleshooting*.
- Latta, Frostburg State: HW-101 restoration, alignment, modifications, HP-23 restoration and schematic pages.
- WB8JKR, *Optimizing the HW-101, SB100-102 Transceivers*; K4JPN, *Improve Your HW-101*; KM5Z refurb summary (pestingers.net).
- Heath list threads: “Problems with HW-101 — a possible solution” (Dec 2020, reversed wafers); “Drifting HW-101” (Jun 2007); “S meter drift” (Feb 2003).

11. Pending manual cross-check

As with the SB-301, I’ll check this plan against the original manual. Items to verify:

- V-number assignments (§2.2).
- The full electrolytic list and voltage ratings (C2/C12/C212/C304/C906 and any others).
- The exact bias procedure and resting-current target.
- The resistance and voltage charts.
- Diode designators and types.
- The manual’s alignment wording and troubleshooting chart.